How-To Library

Informatica

Converting relational input into hierarchical output using Microsoft Azure Data Lake Storage Gen2 Connector

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Abstract

When you want to read data from a relational source and write data to Microsoft Azure Data Lake Storage Gen2, you can use the Hierarchy Builder transformation to convert relational input into hierarchical output. The transformation processes a relational input from the upstream transformation and provides JSON output to the downstream transformation. This article describes how to use the Hierarchy Builder transformation with Microsoft Azure Data Lake Storage Gen2 Connector to convert relational input into hierarchical output.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Data Lake Storage Gen2 Connector

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Overview

When you want to write data from relational sources to Microsoft Azure Data Lake Storage Gen2, you can use the Hierarchy Builder transformation to convert relational input into hierarchical output.

The transformation processes a relational input from the upstream transformation and provides JSON output to the downstream transformation. Use a schema file to create a hierarchical schema that defines the expected hierarchy of the output data. Create and configure the Hierarchy Builder transformation in Data Integration and add it to a mapping. Then, associate the schema definition file with the Hierarchy Builder transformation. When you run the mapping, the Hierarchy Builder transformation converts the relational input into the hierarchical output based on the schema file.

The JSON output generated by the Hierarchy Builder transformation contains data of string data type. Use the Java transformation to convert the data to binary format before you upload data to Microsoft Azure Data Lake Storage Gen2.

To convert relational input into hierarchical output in a Microsoft Azure Data Lake Storage Gen2 mapping, perform the following steps:

1. Configure a JSON schema definition file on the Secure Agent machine.
2. Add a Source transformation.
3. Add a Hierarchy Builder transformation to the mapping.
   - Associate the JSON schema definition file with the Hierarchy Builder transformation.
   - Configure the field mapping and link the relational fields to schema elements that provide the hierarchical output.
4. Add a Java transformation.
5. Add a Target transformation.

6. Run the mapping or the mapping task.

Before you begin

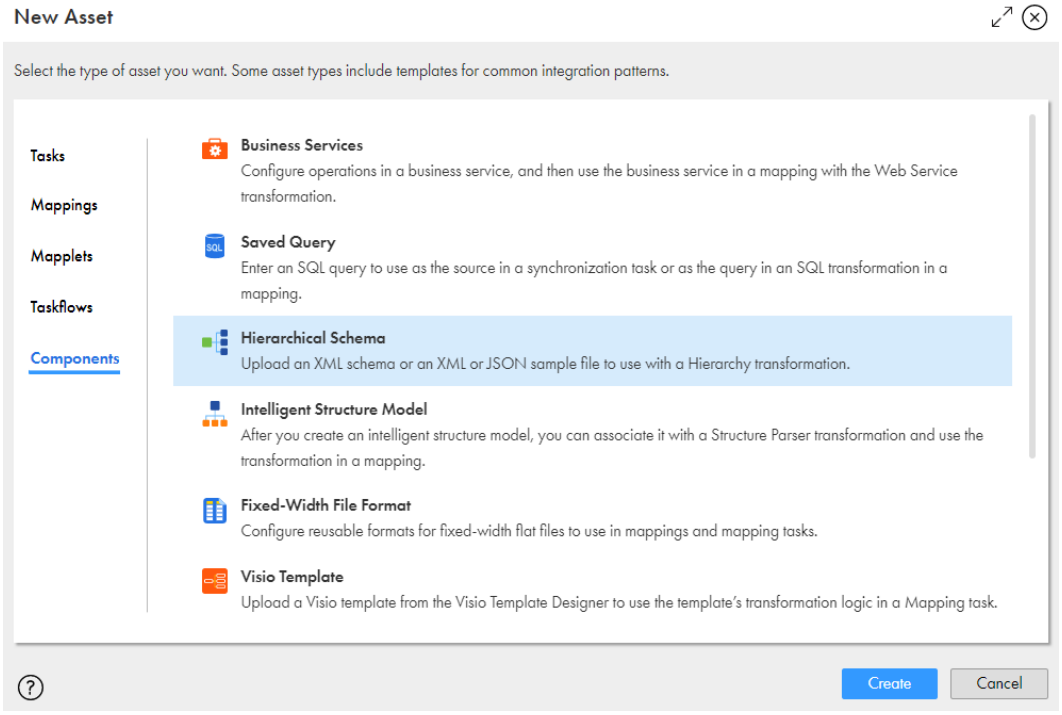
Before you convert the relational input into hierarchical output, complete the following tasks:

- Verify that you have the permission to use the following licenses and packages before you create a mapping:
  - Saas-xmetadataread package
  - DT-XML Processing package
  - UDT for Hierarchy package
  - Data Transformation
  - Java Transformation
  - Heirarchy Builder Transformation
  - Agent Compiler
- Configure the JAVA_HOME environment variable on the Secure Agent machine.

Create a hierarchical schema

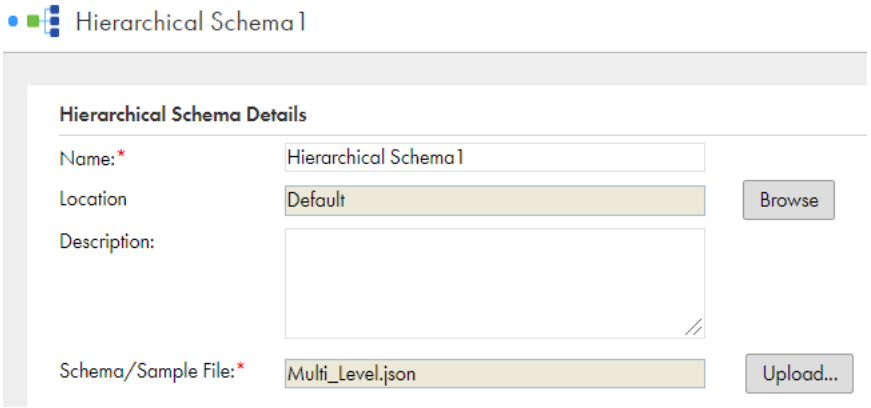
You must associate a JSON schema definition file with the Hierarchy Builder transformation to define the schema of the Microsoft Azure Data Lake Storage Gen2 target object. The Hierarchy Builder transformation converts hierarchical input based on the JSON schema definition file.

1. In Data Integration, click **New** > **Components** > **Hierarchical Schema**.

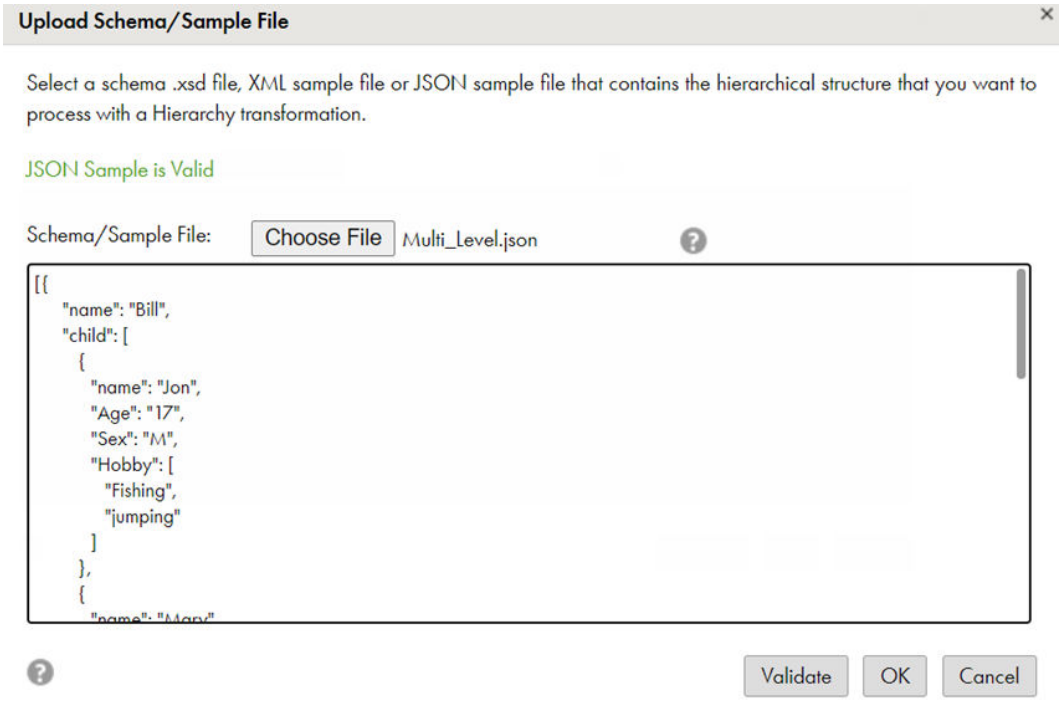


2. Click **Create**.

3. Enter a name for the hierarchical schema and optionally enter a description.



4. To select a schema, click **Upload**.
5. Click **Choose File**, select a sample JSON file, and then click **OK**.

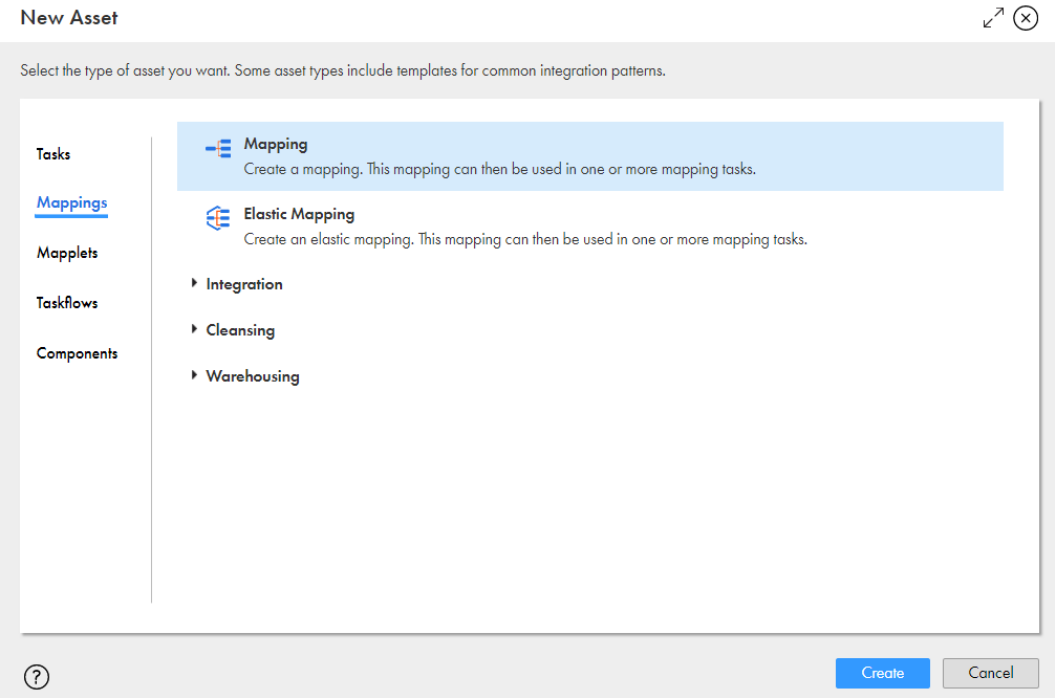


6. Click **Save**.

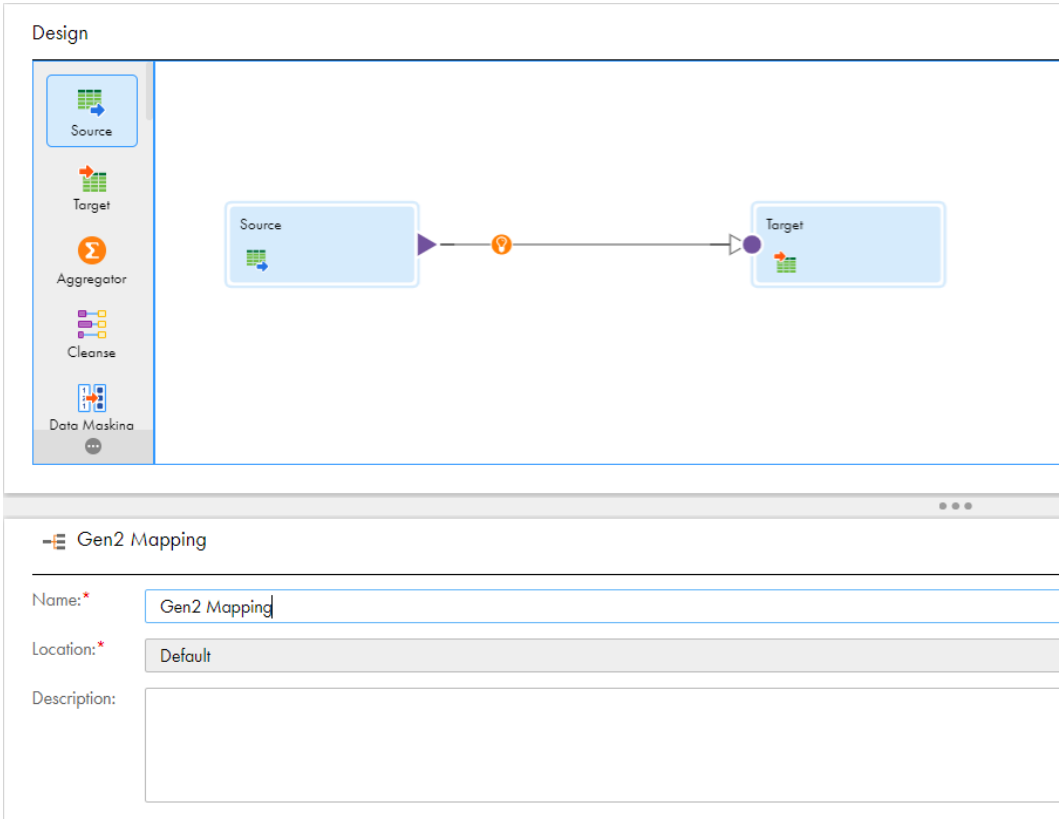
Add a Source transformation

Add a relational source object and configure the source properties.

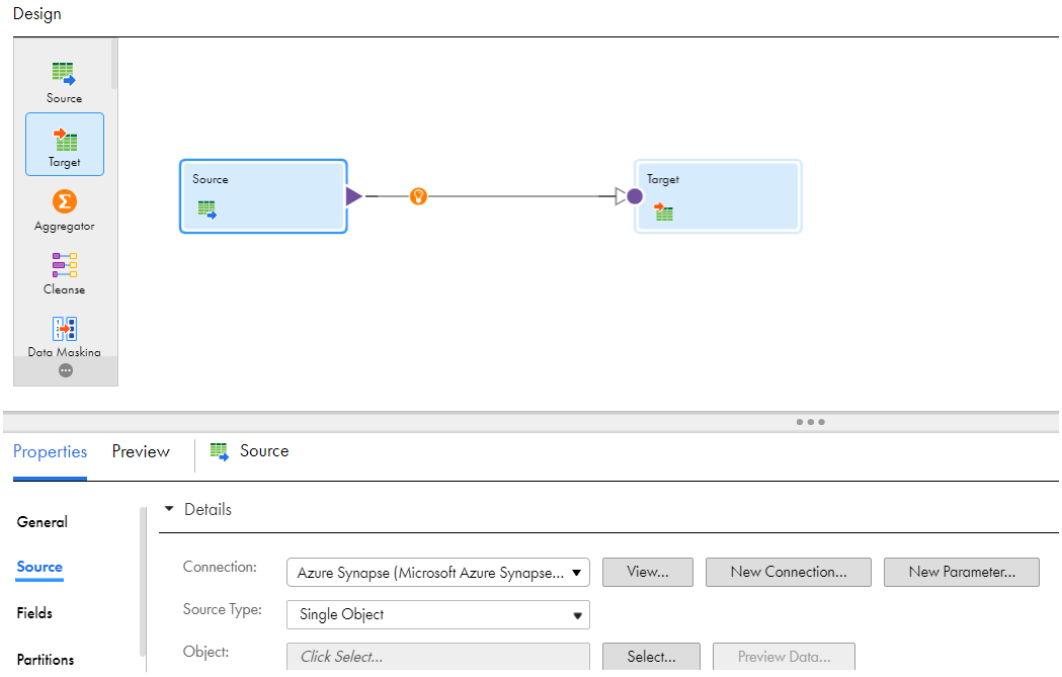
1. To create a mapping, click **New** > **Mapping**.



2. Click **Create**.
   The Mapping Designer appears with a new mapping displayed in the mapping canvas.



3. Enter a name and description for the mapping.
4. Select the Source transformation and configure the properties.
5. On the **General** tab, enter a name and description for the source.
6. On the **Source** tab, select an active source connection and the source object from where you want to read the data.
   In this example we are reading data from a Microsoft Azure Synapse SQL source.



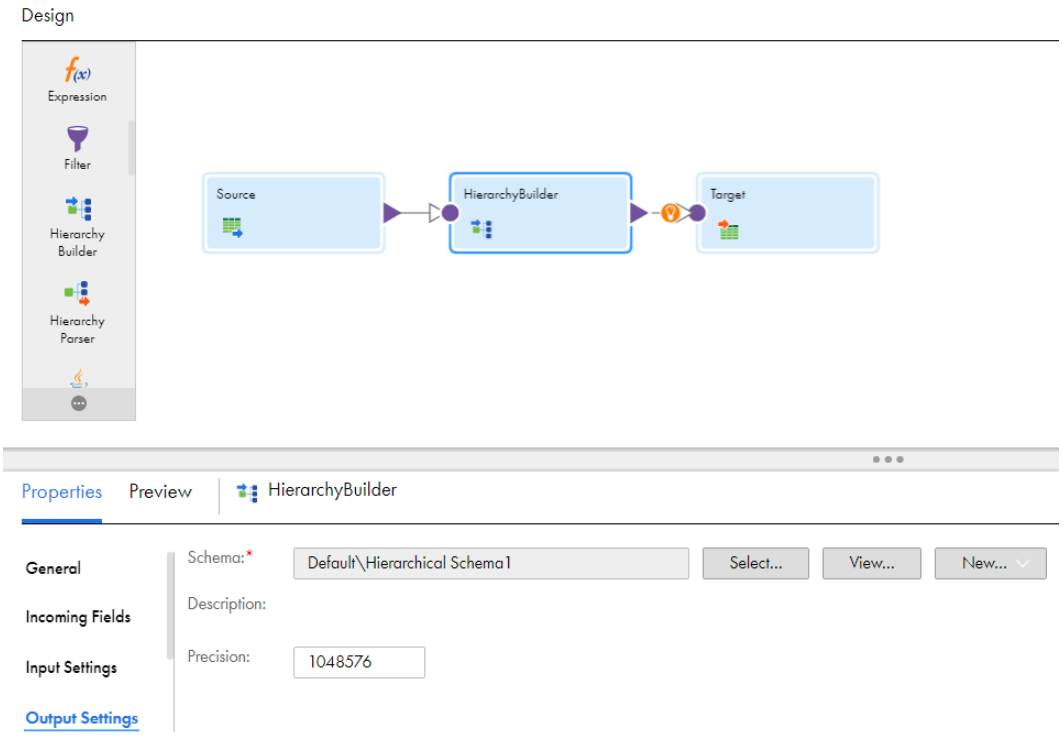
7. Configure the advance properties for the source object, if required.

Add a Hierarchy Builder transformation

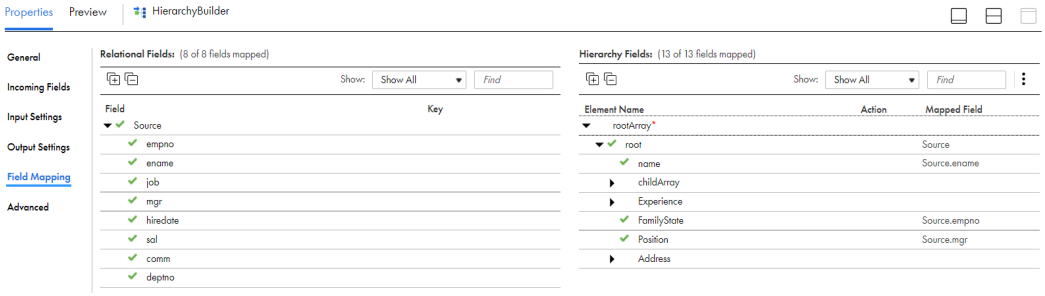
You must add a Hierarchy Builder transformation to the mapping to convert the relational input into hierarchical output. The transformation processes the relational input from the upstream transformation and provides JSON output to the downstream transformation. The Hierarchy Builder transformation produces hierarchical output based on the JSON schema definition file that you associate with the transformation.

1. Add a Hierarchy Builder transformation to the mapping and configure the properties for the transformation.
2. On the **General** tab, enter a name and description for the Hierarchy Builder transformation.

3. Click the **Output Settings** tab and select the schema you created to convert the relational input into the hierarchical output.



4. On the **Field Mapping** tab, configure the field mapping to define which relational elements link to schema elements to provide the hierarchical output. You use the field mapping editor to link relational fields to schema elements. The field mapping editor displays the transformation input elements and output fields. To the left, the editor shows the schema elements. To the right, the editor shows the relational output fields.



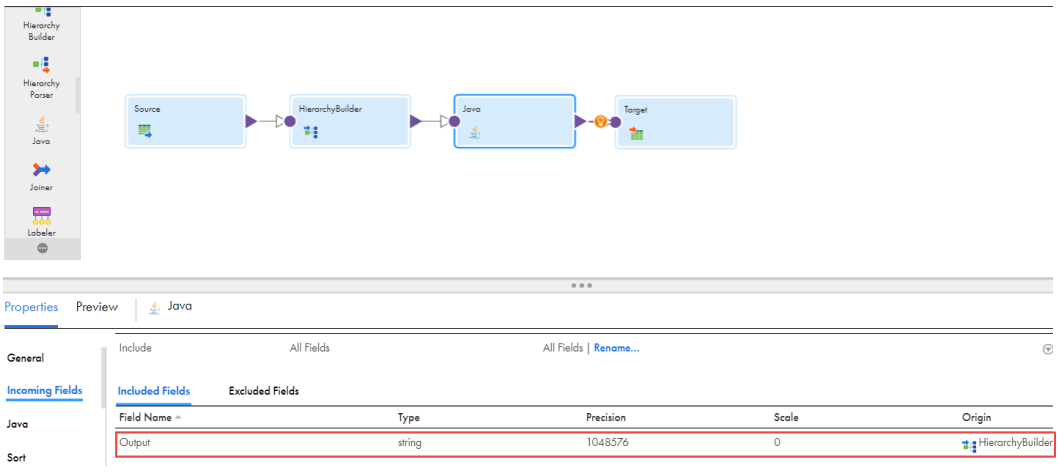
Add a Java transformation

The JSON output generated by the Hierarchy Builder transformation contains data of the string data type. You must use the Java transformation to convert the string data to binary format before you upload data to Microsoft Azure Data Lake Storage Gen2.

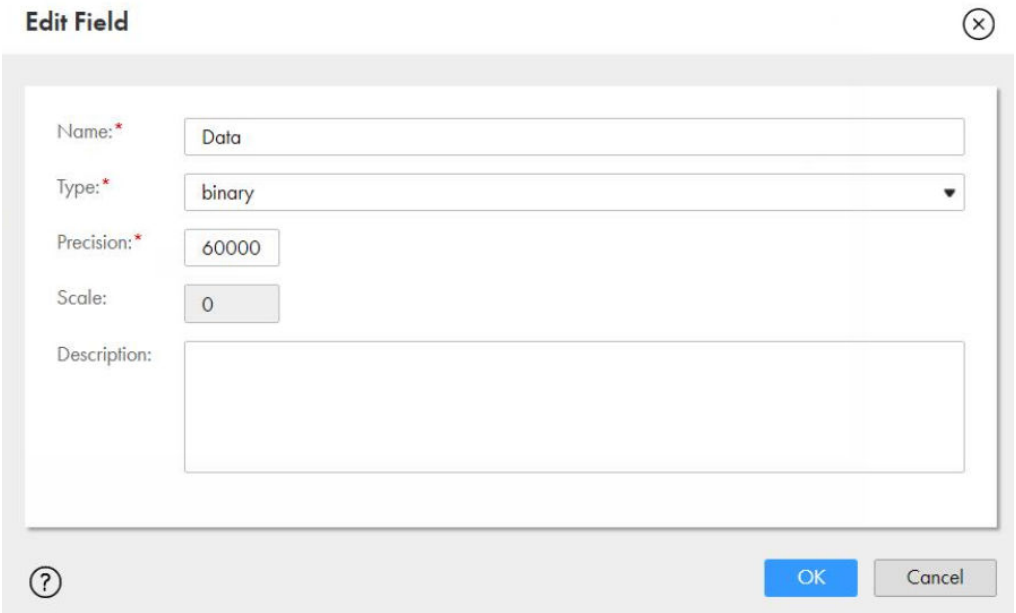
A Java transformation has incoming fields and output fields. Use incoming and output fields as variables in the Java code snippets that you define on the **Java** tab. Incoming fields appear on the **Incoming Fields** tab. By default, the Java transformation inherits all incoming fields from the upstream transformation.

1. Add a Java transformation to the mapping.
2. On the **General** tab, enter a name and description for the Java transformation.

3. On the **Incoming Fields** tab, you can view the incoming fields from the Hierarchy Builder transformation of string data type



4. On the **Output Fields** tab, add output fields for the output data that you want to pass to the downstream transformation.
   a. Click **Add Field.**
   b. Enter a name for the field.



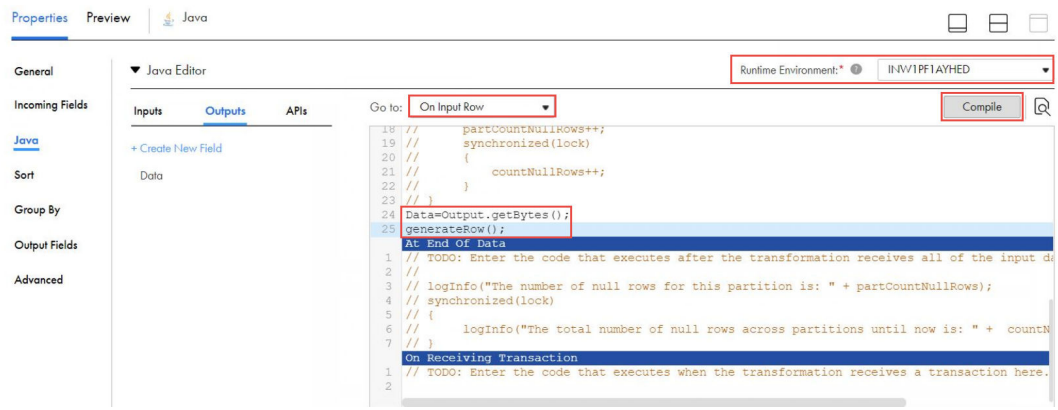
   c. In the **Type** field, select **Binary**.
   d. Specify the precision, and click **OK**.
5. On the **Java** tab, enter the Java code snippet in the **On Input Row** section in the following format:

```
<output field>=<input field>.getBytes();
generateRow();
```

For example,

```
Data=Output.getBytes();
generateRow();
```

The following image shows the Java code snippet in the Java editor:

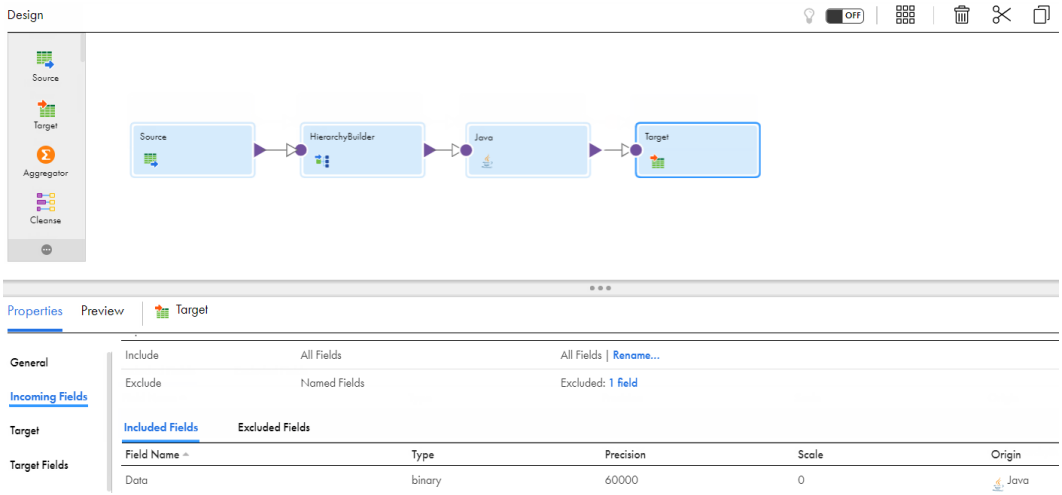


6. Select the **Runtime Environment**.
7. Click **Compile**.

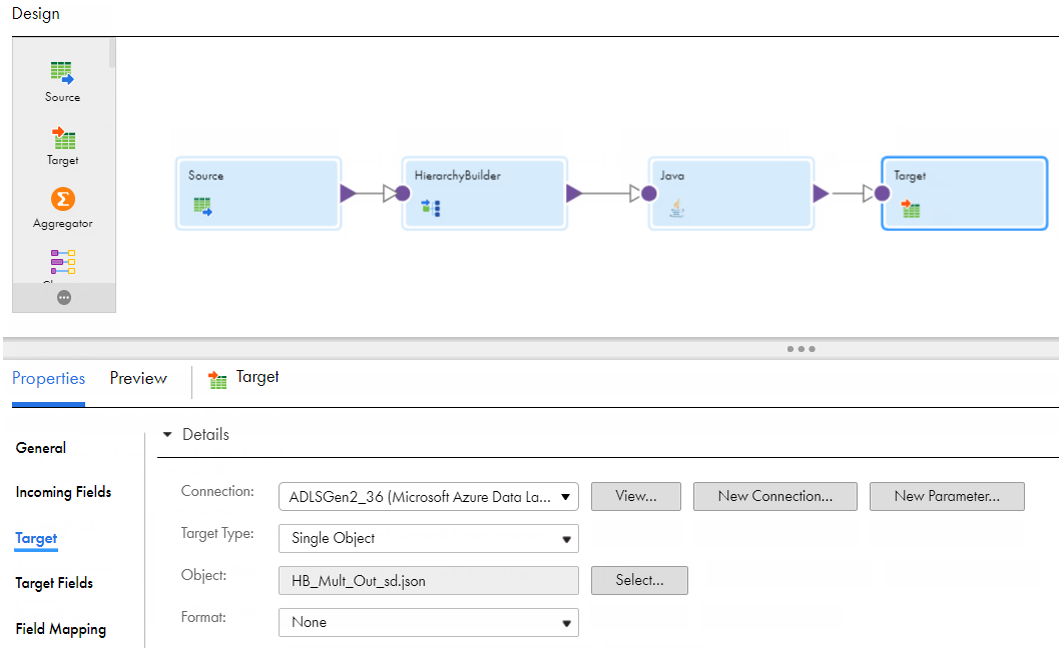
Add a Target transformation

Use the Target transformation to define the Microsoft Azure Data Lake Storage Gen2 target in the mapping.

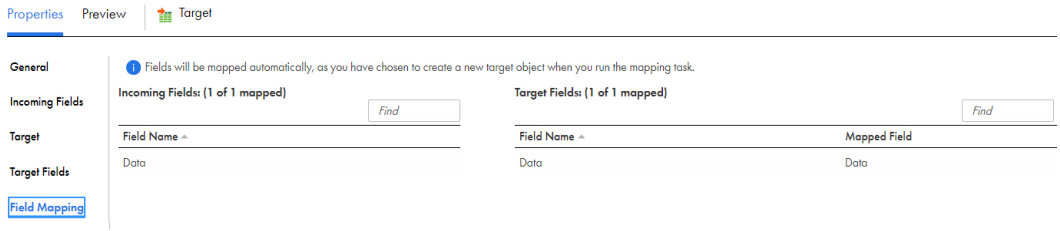
1. Connect the Hierarchy Builder transformation to the Target transformation.
2. Select the Target transformation and configure the properties.
3. On the **General** tab, enter a target name and optional description.
4. On the **Incoming Fields** tab, you can view the incoming fields of binary type from the Java transformation.



5. On the **Target** tab, select an active Microsoft Azure Data Lake Storage Gen2 connection and select a Microsoft Azure Data Lake Storage Gen2 target object.



6. In the **Format** field, select **None**.
7. On the **Field Mapping** tab, map the incoming field from the Java transformation to the target field.

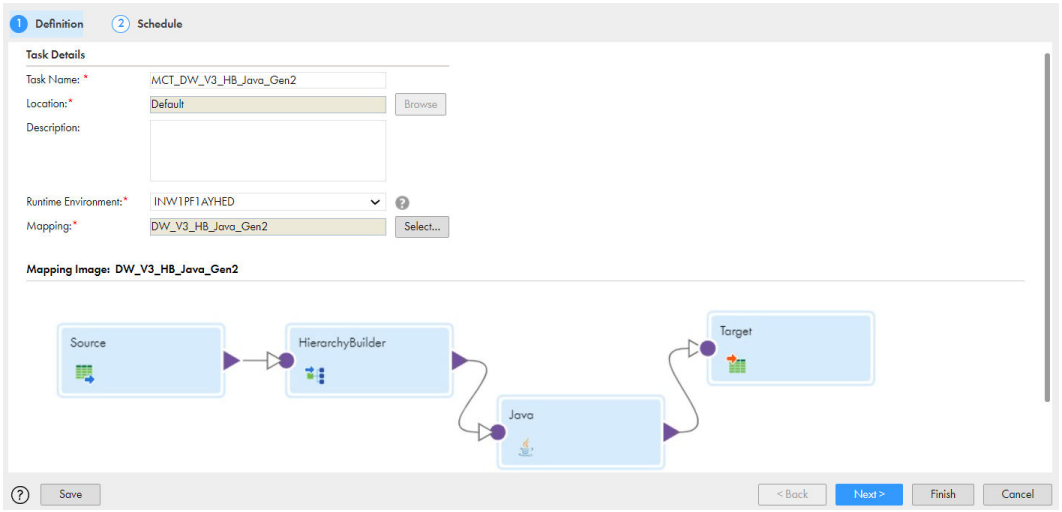


8. To save the mapping, click **Save**.

Configure a mapping task

You can save and run the mapping directly or configure a mapping task, add the configured mapping, and run the mapping task.

1. To create a mapping task, click **New** > **Tasks**, and then click **Create**.
2. In the New Mapping Task window, provide a task name, and provide the configuration details.



3. Select the configured mapping, and click **OK**.
4. Configure a schedule and advanced options, if required.
5. Click **Finish**.
6. Click **Save**, and then click **Run**.

Author

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